



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL SCIENCE [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 5 printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

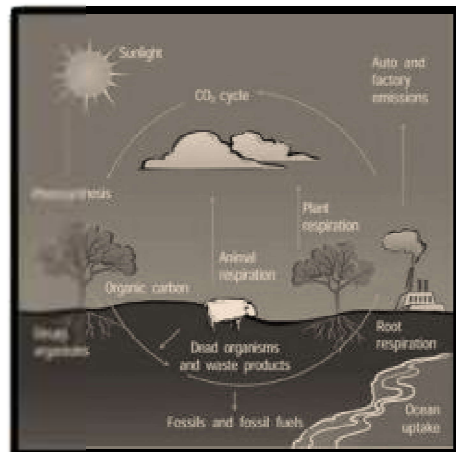
Choose the correct answers to the questions from the given options.

[16]

(Do not copy the questions, write the correct answers only)

- (i) Energy plantation: Fast growing crops:: _____ : Drought resistant crops
- (a) Strip farming
 - (b) Monocropping
 - (c) Dry farming
 - (d) Crop rotation
- (ii) Which of the following restricts the use of drip irrigation?
- (a) Clogged dippers
 - (b) Salting increases
 - (c) Initial cost of drip irrigation system is very high
 - (d) Too much water is wasted
- (iii) Assertion: Three quarters of the world's population is inadequately fed so there is hunger and malnutrition in the developing countries.
Reason: The factors of food shortage in the developing countries are poverty, unsuccessful agriculture and unequal distribution in food supply
- (a) A is true and R is the correct reason for A.
 - (b) A is true but R is not the correct reason for A.
 - (c) A is true but R is false.
 - (d) A is false but R is true.
- (iv) Death rate is also known as
- (a) Demography
 - (b) Natality
 - (c) Mortality
 - (d) Fertility

- (v) Rearrange the following steps in the ascending order that trace the rate of solid waste management:
 Step 1: Transportation of waste
 Step 2: Collection of waste
 Step 3: Waste disposal
 Step 4: Segregation of waste
- Step 1, Step 2, Step 3, Step 4
 - Step 3, Step 2, Step 1, Step 4
 - Step 1, Step 3, Step 2, Step 4
 - Step 3, Step 1, Step 2, Step 4
- (vi) The aim of land reforms is:
- Escalating poverty
 - Increasing agriculture productivity
 - Declining co operative farming
 - Encouraging land fragmentation
- (vii) Green manure acts as a solidifying matter that:
- Increases alkalinity of the soil
 - Decreases alkalinity of soil
 - Increases acidity of soil
 - Decreases acidity of soil
- (viii) The benefit of satellite image is:
- It has a permanent record
 - The data is difficult to collect
 - The data is processed very slowly by the computer
 - It causes several environmental issues
- (ix) Which of the following is an example of Ex situ conservation?
- Kanha National Park
 - Sunderban Biosphere Reserve
 - Bharatpur Bird Sanctuary
 - Nandankanan Zoo
- (x) Which function of biodiversity is shown in the picture ?



- (a) Economical
- (b) Ecological
- (c) Physiological
- (d) Social

(xi) Pick out the incorrect pair from the following:

- (a) Solar Energy- Photovoltaic cell
- (b) Wind Energy- Wind Mills
- (c) Hydro Energy- Wells
- (d) Geothermal Energy – Turbines

(xii) To curb pollution, Delhi bus transport system has become the biggest CNG run transport system in the world.

The main reason for using CNG is:

- (a) Easy availability
- (b) More efficiency and less polluting
- (c) Gives less mileage
- (d) CNG fuel tanks are bulky

(xiii) Ram performed a waste audit in his local community and found that it could pose a risk to human health.

Which of the following four tenets (4R) did he implement to manage the waste?

- (a) Rewrite, Reuse, Reduce, Rewarm
- (b) Rejoice, Reduce, Rewrite, Reuse
- (c) Rewarm, Recycle, Refuse, Rejoice
- (d) Refuse, Reduce, Recycle, Reuse

(xiv) Match the following columns correctly:

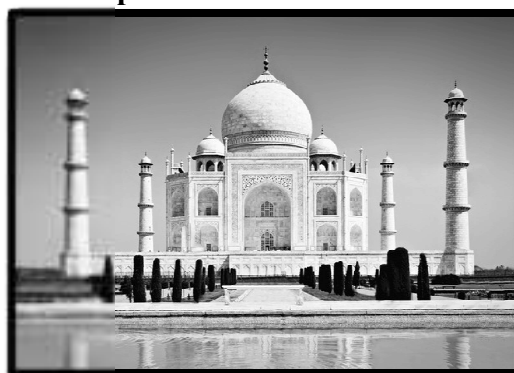
Column A

- (A) Dodo
- (B) ESP
- (C) CITES
- (D) Population control
- (a) A-i, B-iii, C-ii, D-iv
- (b) A-ii, B-iv, C-i, D-iii
- (c) A-ii, B-iii, C-iv, D-i
- (d) A-iii, B-iv, C-ii, D-i

Column B

- (i) Contraceptives
- (ii) Animal trade
- (iii) Extinct
- (iv) Electrodes

(xv) Which environmental phenomenon is responsible for the degradation of the monument shown in the picture?



- (a) Smog
- (b) Acid rain
- (c) Global warming
- (d) Sprawls

(xvi) Integrated Rural Development Programme is a self employment programme of the government of India to raise income generation capacity of the poor.

Which scheme is undertaken under the programme?

- (a) Urban Sanitation Programme
- (b) International Food for Work Programme
- (c) Training of pensioners for self employment
- (d) Development of women and children in Rural areas.

Question 2

- (i) State two reasons for recycling water. [2]
- (ii) Mention any two advantages of landfill. [2]
- (iii) How does women empowerment check population growth? [2]
- (iv) Define allelopathy. How is it useful to farmers? [2]

Question 3

- (i) List two ways in which human lifestyle directly degrades nature. [2]
- (ii) Explain the role of gene bank in agricultural research. [2]
- (iii) Mention any two objectives of CBD. [2]
- (iv) What is meant by extinct in wild? Give an example. [2]

Question 4

- (i) Describe sewage sludge. How is it important for plant growth? [2]
- (ii) Write any two characteristics of secondary cities. [2]
- (iii) Write the full form of INSP. Why is it useful to farmers? [2]
- (iv) What does the picture depict? How can this be prevented? [2]



SECTION B (40 MARKS)

(Attempt any four questions from this section)

Question 5

- (i) In the past two decades, genetically modified organisms (GMOs) have become increasingly common in agriculture. These are plants or animals whose DNA has been altered using modern biotechnology to express desirable traits—such as resistance to pests, tolerance to drought, or [5]

faster growth. In India, crops like Bt cotton have been widely adopted by farmers because of their ability to kill certain pests without the use of chemical pesticides.

Do you think use of GMOs should be continued? Justify your answer.

- (ii) *Man must realize that in order to survive, he must handle nature with care.* [5]
Based on the statement given above, analyze any five ways in which humans are responsible for the loss of biodiversity.
understanding

Question 6

- (i) What is conservation tillage? Mention any three advantages of Conservation Tillage. [5]
- (ii) Discuss any five strategies to control vehicular pollution. [5]

Question 7

- (i) [5]



Which project was undertaken to protect the animal shown in the picture?
Enumerate the achievements of the movement.

- (ii) How do national parks differ from biosphere reserves? [5]

Question 8

- (i) Suman grows vegetables like tomatoes and brinjals in her 2-acre farm. [5]
She used to rely heavily on chemical pesticides to control insects and worms. After a few seasons, she noticed that her produce was being rejected in the local market due to pesticide residue. Also, some pests were no longer responding to the sprays. She contacted an agricultural officer who advised her to follow IPM. Over time, her vegetables became healthier, the soil recovered, and her profits improved.
What is IPM? How did it help Suman improve her produce?
- (ii) Define sustainable development. Write four ways of making your city sustainable. [5]

Question 9

- (i) Enumerate the effects of urbanization. [5]

- (ii) Mention the achievements of Ralegaon Sidhhi. [5]

Question 10

- (i) In the semi-arid region of Rajasthan, frequent droughts and overuse of firewood had degraded the land and increased energy scarcity. The local forest department, in partnership with an NGO, started an Energy Plantation Project. They selected fast-growing, drought-resistant species like *Jatropha*, *Eucalyptus*, and *Casuarina* that could thrive on poor soils and be used as biofuel, firewood, or for charcoal production. Explain the term energy plantation. How did it benefit the people of the semi arid region? [5]

- (ii) [5]



Identify the convention. What is its purpose? Why is it important?